

Dynamic Maze Solver Comparison

HTW EB-JEPA

20 June 2026

Scope

This document summarizes the dynamic-maze experiments: symbolic A* baselines, A*-distilled JEPA subgoal controllers, and A*-free world-model controllers. Animated GIFs are saved next to this report; the PDF contains static start/end previews and clickable links to the GIF files.

The table covers 768 closed-loop evaluation episodes across 12 methods. Most rows use 64 episodes. The symbolic/A*-distilled and bootstrap runs come from different launchers, so oracle-normalized ratios should be read as operational comparisons rather than as one perfectly matched benchmark.

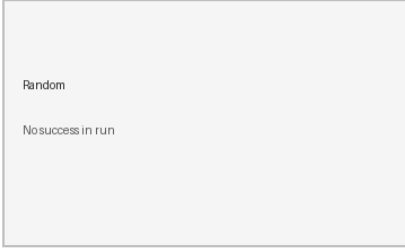
Metric Definitions

- Success is the empirical probability of reaching the goal over 64 episodes.
- Runs is the number of closed-loop evaluation episodes available for that method.
- Steps is the mean number of executed moves.
- Blocked moves counts attempted moves into walls/closed doors.
- A* efficiency is the evaluator’s per-episode $A_0 / \max(\text{steps}, A_0)$, when available.
- Step/oracle is mean oracle steps/mean solver steps.
- Combined is success times Step/oracle; it rewards both reaching the goal and doing so efficiently.

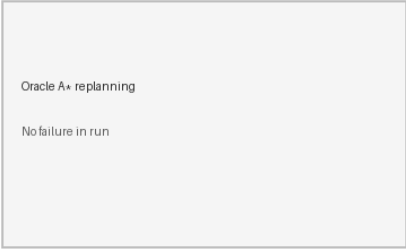
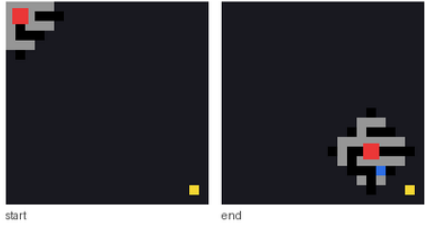
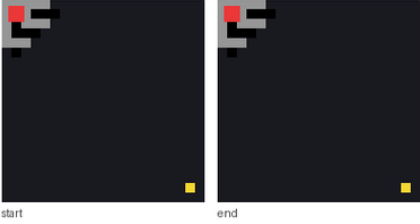
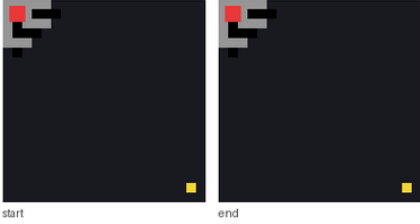
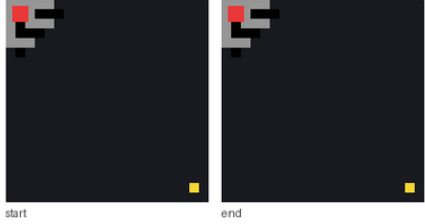
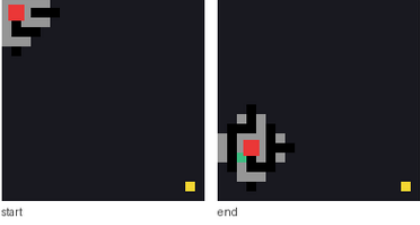
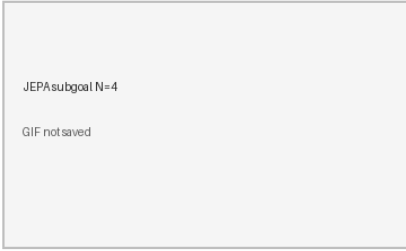
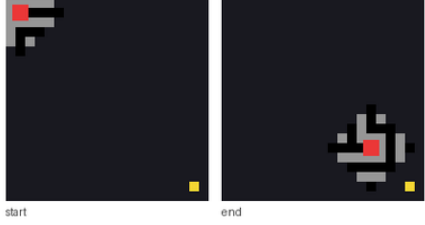
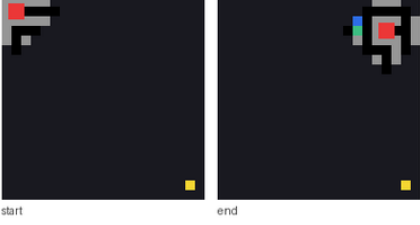
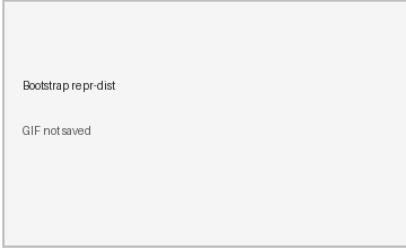
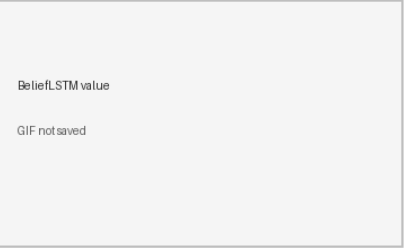
Results

Method	Runs	Success	Steps	Blocked	A* eff.	Step/oracle	Combined
Oracle A* replanning	64	100.0%	70.7	0.0	0.966	100.0%	100.0%
Memory optimistic A*	64	100.0%	79.7	0.0	0.909	88.7%	88.7%
Fog optimistic A*	64	14.1%	437.7	0.0	0.134	16.2%	2.3%
Fog conservative A*	64	0.0%	488.0	0.0	0.000	14.5%	0.0%
Memory conservative A*	64	0.0%	488.0	0.0	0.000	14.5%	0.0%
Random	64	0.0%	488.0	247.5	0.000	14.5%	0.0%
JEPA subgoal N=2	64	57.8%	280.8	103.3	0.403	25.2%	14.6%
JEPA subgoal N=4	64	73.4%	230.7	93.0	0.494	30.6%	22.5%
Bootstrap learned value	64	70.3%	289.5	88.6	—	23.3%	16.4%
Bootstrap probe-pos	64	65.6%	323.2	84.7	—	20.9%	13.7%
Bootstrap repr-dist	64	75.0%	244.9	85.1	—	27.6%	20.7%
BeliefLSTM value	64	73.4%	271.8	100.5	—	24.8%	18.2%

Success Examples

Oracle A* replanning success=100.0% steps=70.7 Oracle A* replanning - positive  start end	Memory optimistic A* success=100.0% steps=79.7 Memory optimistic A* - positive  start end	Fog optimistic A* success=14.1% steps=437.7 Fog optimistic A* - positive  start end
Fog conservative A* success=0.0% steps=488.0 Fog conservative A* No success in run 	Memory conservative A* success=0.0% steps=488.0 Memory conservative A* No success in run 	Random success=0.0% steps=488.0 Random No success in run 
JEPAsubgoal N=2 success=57.8% steps=280.8 JEPAsubgoal N=2 - positive  start end	JEPAsubgoal N=4 success=73.4% steps=230.7 JEPAsubgoal N=4 - positive  start end	Bootstrap learned value success=70.3% steps=289.5 Bootstrap learned value - positive  start end
Bootstrap probe-pos success=65.6% steps=323.2 Bootstrap probe-pos - positive  start end	Bootstrap repr-dist success=75.0% steps=244.9 Bootstrap repr-dist - positive  start end	BeliefLSTM value success=73.4% steps=271.8 BeliefLSTM value - positive  start end

Failure Examples

Oracle A* replanning success=100.0% steps=70.7 	Memory optimistic A* success=100.0% steps=79.7 	Fog optimistic A* success=14.1% steps=437.7 Fog optimistic A* - negative  start end
Fog conservative A* success=0.0% steps=488.0 Fog conservative A* - negative  start end	Memory conservative A* success=0.0% steps=488.0 Memory conservative A* - negative  start end	Random success=0.0% steps=488.0 Random - negative  start end
JEPAsubgoal N=2 success=57.8% steps=280.8 JEPAsubgoal N=2 - negative  start end	JEPAsubgoal N=4 success=73.4% steps=230.7 	Bootstrap learned value success=70.3% steps=289.5 Bootstrap learned value - negative  start end
Bootstrap probe-pos success=65.6% steps=323.2 Bootstrap probe-pos - negative  start end	Bootstrap repr-dist success=75.0% steps=244.9 	BeliefLSTM value success=73.4% steps=271.8 

GIF Links

Method	Success	Failure
Oracle A* replanning	success GIF	—
Memory optimistic A*	success GIF	—
Fog optimistic A*	success GIF	failure GIF
Fog conservative A*	—	failure GIF
Memory conservative A*	—	failure GIF
Random	—	failure GIF
JEPA subgoal N=2	success GIF	failure GIF
JEPA subgoal N=4	success GIF	—
Bootstrap learned value	success GIF	failure GIF
Bootstrap probe-pos	success GIF	failure GIF

Bootstrap repr-dist	success GIF	–
BeliefLSTM value	success GIF	–

Notes

- **Oracle A* replanning:** Full current grid A* replanning. Strong myopic oracle, not stochastic-optimal.
- **Memory optimistic A*:** A* on accumulated visible map, unknown cells treated as free.
- **Fog optimistic A*:** A* on current local observation, unknown cells treated as free.
- **Fog conservative A*:** A* on current local observation, unknown cells treated as walls.
- **Memory conservative A*:** A* on accumulated map, never-seen cells treated as walls.
- **Random:** Uniform random cardinal action baseline.
- **JEPA subgoal N=2:** A*-distilled waypoint head; A* not queried at eval.
- **JEPA subgoal N=4:** A*-distilled longer waypoint head; first saved GIFs are all successes.
- **Bootstrap learned value:** A*-free local exploration + hindsight TD value.
- **Bootstrap probe-pos:** World-model rollout scored by decoded position distance.
- **Bootstrap repr-dist:** World-model rollout scored by latent distance to goal.
- **BeliefLSTM value:** Frozen JEPA + recurrent belief state + learned value.

Main Reading

The strongest non-oracle controller in this batch is still representation distance in JEPA latent space: it reaches 75.0% success with 244.9 mean steps. The BeliefLSTM value improves the first learned value head, reaching 73.4% success, but it still makes more blocked moves. The next useful Track 7 iteration should make the learned value explicitly penalize blocked transitions and use memory signals that distinguish seen-passable, seen-blocked, and unseen cells.